THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 5: **Exploring Forces** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A skydiver falls at constant speed with the parachute open. The forces on her are: 1

(a) zero **(b)** unbalanced downward
(c) unbalanced upward **(d)** balanced

2. A cyclist stops pedalling and slows down because of: 1

(a) weight only **(b)** gravity pulling forwards
(c) friction and air resistance **(d)** the magnet

3. Fish and boats have streamlined shapes to reduce: 1

(a) weight **(b)** fluid friction (drag)
(c) mass **(d)** gravity

4. Which has the largest force of friction? 1

(a) a heavy box on a rough floor **(b)** a light box on a smooth floor
(c) a box on ice **(d)** a box on oiled floor

5. The weight of an object on the Moon compared to Earth is: 1

(a) more **(b)** less **(c)** same **(d)** zero

6. The brake pads of a bicycle are made of rubber to: 1

(a) increase friction with the rim **(b)** cool the wheel
(c) decrease friction **(d)** make it light

7. Static charges can be dangerous near: 1

(a) paper **(b)** stones
(c) water **(d)** petrol or gas leaks

8. The mass of a stone is 2 kg on Earth. On the Moon its mass is: 1

(a) 0 kg **(b)** 2 kg **(c)** more than 2 kg **(d)** less than 2 kg

9. A wheelbarrow with a tyre is easier to move than a sledge because: 1

(a) rolling friction is less than sliding friction
(b) wheels are heavy
(c) gravity differs
(d) sledges are magnetic

10. A person standing in an open field during a thunderstorm should: 1

(a) use a mobile phone **(b)** stand under a tall tree
(c) crouch low and keep feet together **(d)** hold an umbrella high

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why does a charged body lose its charge slowly in humid air? 2

12. Explain why a plastic comb attracts paper bits only after rubbing. 2

13. A weight of a body is 60 N on the Earth. If the Moon's gravity is one-sixth, find the weight on the Moon. 2

14. State two differences between friction and gravitational force. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain the factors that affect the friction between two surfaces. 3

16. Explain how force changes the state of motion of a body with an example of each effect. 3

17. Explain why a thin magnetised needle floats in a bowl of water on a cork and aligns itself. 3

OR

Why does a parachute fall slowly? Use the idea of force. 3

18. What do you mean by tug of war being a balanced situation? Explain it when one team wins. 3

OR

Differentiate between a magnetic force and an electrostatic force with two points. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A box of weight 40 N is pushed along the ground with 15 N and friction is 15 N. Describe the motion. What if the push is raised to 25 N? 5

OR

Explain why lightning is more likely to strike tall objects and list five safety measures. 5

20. Describe with a labelled description how an electroscope shows the presence of charge and how it can be discharged. 5

OR

Describe an experiment to show the force of gravity acts on all objects regardless of mass in the absence of air resistance, and give the conclusion. 5

END OF QUESTION PAPER — Total: 40 Marks

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